

Search sheets: how to run the activity

LL Ms Un plug ged

The pages after this one are 36 **search sheets**, built from *The Cat in the Hat* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

If the room is a different size, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 54–55 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **in the**, the boxed token is the previous token and **the** is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 8 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *the*? it’s a purple one” — and most of the page stops being worth reading. The swatches are chosen to answer to these words, so the words are the ones to use; the sheets themselves print no key, because a strip of swatches at the top of a page of swatches is one more thing to scan past.

black grey red brown green blue purple magenta

Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Quiet rounds. One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

1962 tokens, 240 unique · 2.83 bits/token · perplexity 7.1 · branching factor 4.32

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token and its next token on the board.

Round 1. Call out the last token on the board (**the**). Another sheet has a match, so that person raises a hand and reads out *milk*.

Round 2. Call out the last token on the board (**milk**). Another sheet has a match, so that person raises a hand and reads out *on*.

Keep going for as long as you like. The text the room writes will sound like *The Cat in the Hat* without ever repeating it — that is the model generating, not the corpus being read out.

the token pair that answers on the board

in the in the

the milk in the milk

milk on in the milk on

in the game that fish in said ,
 things will Why do what would . that
 kites on tell me like , did not
 the fish is not can have so ,
 a pot fly kites find this I will
 the ball tip of sad kind you some
 fan ! us this , wet ! so
 have to do as should not ! look
 wet to had to saw them with the
 and I bump on . have I do
 he said we do that is milk on
 the way . and then I and so
 to you ! then not mind , I
 will show on a . not

find us did you Why , not all
 ! and and Thing day . and the
 so tame , white said the hands with
 a cake , what ! said is a
 shame ! wet day kites , I call
 . . your with my then we
 can hold put them I do . your
 we had have Lots . he tip of
 to us the milk will not saw him
 you do the cup mess , on the
 fish , like to he put fan as
 do something house and ! said
 that is the fish milk and him !
 what to call up about it things and

then , him step mat ! fan ,
 put me . you what will house .
 ? Oh you can call them on a
 I said deep and wet , the cat
 ! and can hold to us . we
 and Sally have any ! Thump hop up
 with a things I ! Who , I
 said the saw him is not the cat
 the books mess is , I toy ship
 and I I know we saw . and
 he said like it hands . milk ,
 not a will see . he about with
 that are day . shame ! fish !
 so so bad tricks do !

what to him pick ! said is not
 the hat . Yes fish . two
 house ! call this with a , with
 are pink said the want to toy man
 so all put them shook their hands with
 you ? will like ! look fan with
 bad , day . wet and like these
 can hold ? Well shame ! mess !
 I will ! so on there he fell
 we did and so do ? about .
 not know that is then Sally . that
 the hat saw one to find and big
 things fall . do hop on , said
 I will the cake deep in

! what things they our fish get them
gown with fish in then Sally . you
wet day house ! lit . ! and
the toy toy ship not sunny to us
want to day ? I can some good
we looked he gave on his that bump
would you what a and Sally . so
put me so . do not I say
! now good fun will show saw him
the cat Think of , said a ball
can hold said to here . like it
with a the house shook up , no
you should ? Why and we are two
. there like . Thing

. then our fish ! and good ,
on a house ? and down . I
Sally . shook with Lots of get out
what the the hat a cup can hold
things yet some fun are good there with
fish as not all the cat he said
so we ? tell toy ship ! I
. But will not . our From up
and Thing to shake would she , and
I do , we put them I wish
you hear like to said the yet !
then I gown ! that . we saw
the hall kind of now you ! bump
here When not be at me

Sally and pick up that is ! and
 , how here When so big you sank
 their two things in like it hops and
 are tame good games not like .
 what did , and said the get those
 then those fish said ! sit they will
 house ! our toy not all Lots of
 to be the cat we saw .
 ? and a cup he came some new
 . I on the and Thing ! he
 . and I like there , would you
 the cat at all gown , now look
 bit ! of her can hold I said
 and this will she the hat

has gone then , here is things that
 and the he was I will a bow
 could see there that we had so .
 will hit be in at this not like
 I will . what . my ! then
 can get that is ! we they will
 their kites bit ! on the our mother
 said the not be ! you some milk
 are not fish said pick up and the
 ? I like that good tricks what to
 now ! you wish . I , wet
 the fish of the get rid the cat
 house ! to get , too bite you
 . we Sally and the cake

tail I that is good trick he lit
gave them things . like to dish !
of this be here cold to can fan
! then and the PLOP ! now !
what you on my not . , not
here to they play we saw . .
? what at all ! said then our
not shine house ! a pat hear !
our mother to fall there on you up
could , the rake the cat said the
fish . pick it I can I will
. he so we . But , said
. and Sally and will have For the
bit . ! those the fan

, out he got to do into a
hear ? ! the a look , then
! and and I our fish fish shook
! what were down pick up dish ,
things . I will not in could do
Sally and there is the hat be here
here . at me on the the fish
they like ? should in the will show
how do fast as now ! house all
. she that were of this . .
cold , I do not fly the hook
said , pack up can get ! Oh
like this what will good game . .
we can then he you pack

we saw a rake fast as of good
 in the can hop will not he
 be here kicks and not be books!
 up and ! I on the man!
 and all , she things ran I can
 could play our fish kinds of give you
 , what that I cold , we
 fish . you sit ! I cat go
 . and string of house For how that
 what would I hop the cat the dots
 ? Oh good things said to to know
 then something now , here When
 he shut . at that Sally and
 like the the house ! look they like

said to can have . these be here
 , we things they ? now one and
 in with two things the day what a
 and the not want how I on the
 at me will give strings , to the
 a book then he ! so and Thing
 ! if box , I , now look
 playthings and books , like it
 up these of something the pot
 mother's new ! they . then
 . and our house fish in . I
 we saw net . wood box fast .
 , so you like that I lot of
 I can show you cat in the ball
 they are good fun

I call show to I can box are
 like our one little the house our new
 a tip , said ! said to say
 you another trick that little toy , the
 ! her now , net . in the
 ! put out of , Why . this
 cat in lot ! fish said what a
 of good . it . and they should
 pat . two books . Oh good game
 things that said the not be we sat
 how . will not the box play .
 books ! at all be here and kicks
 then he fast as that day and so
 ? would up the the fish

, with . so . I to the
 the things then he trick , fast !
 out and we can net , last !
 will come looked ! they should the hat
 Thing one and a Oh , two things
 ! do I can fish came pot !
 tricks . one and the fish cat in
 said the . a in the , do
 a box our mother and look now .
 at last ! and be about box .
 I do you have ! said that is
 up on , said show them not one
 came down play ! no fear little fish
 . this ? Oh of look

no , I I said show you
up those box with you do we sat
. have pot . will bump the cake
then our the mat two , Oh ,
that they out ! little bit not wish
Thing two . the ship and red
looked ! now look one one ! at
, no said the of his in the
, and and I net I funny !
sat in , here they will cat .
came in our fish at me ! those
hall . ! put sun did a bump
fish . play , tricks . to go
. they I can ? and

of Thing Oh dear and the . there
our fish box . in the . I
little bit ! how ! then two .
them go ! then not . one and
. with cat in said , play ball
pot . , what tricks , no fear
Thing two bow . and you hall .
out ! take them will find the cat
we sat this mess now ! came down
sun is you get they want , the
up up look at fish on I can
if your book one at this the wall
the sun . that I let a lot
, I to do sat there

, jumps not pick . you our game
look ! if you to do and the
in a and a take a fall .
When your of one , how do not
we looked they ran play . ! said
the hook the cat little bit two things
fear . . I the milk at all
this mess said the came down , look
a big them a no fun I do
out of ! have home ! I saw
box came ! I Oh , one hand
up high pot . run down sat there
Thing one his hat cat . tricks are
you will fish to . it

up . at me ! they cat .
When our in the the cat came Thing
, now of the hand ! the cat
and Thing . . and with Thing
to the , I sat in no fear
you now we tell Oh , the wall
those things I call new tricks play .
do ? they have look at asked you
one ! his hat I always said the
fear ! them out , said . no
out ! fall ! Thing one if I
not know ! then little toy a little
pot . ! said bed ! two things
and I this game

Thing one new . at this a tip
 came back this way Well . we saw
 the house , and out of not be
 . they When your said , cat .
 went on and Thing his head , said
 and they I had fall ! look at
 no fear ! I fear ! ! now
 . I . in is what ! he
 Make that one ! up all the cat
 with a you do of my do .
 Oh , tame . was gone thumps and
 them out the hat shake hands in the
 I can fell on they ran those things
 if mother two and

cat with said the out . a fish
 ! put . . said our tame !
 those things ! and I bet went away
 cake ! I know Make them Oh ,
 another good it was they should
 new rake up all fear , shake hands
 Thing one know how no ! , that
 the fish and the fell into . I
 wall ! with big not know do not
 . you look at this house and you
 the things ! and of his in the
 , said them I one little two !
 is not the fish was out mind at
 you did say . we could

! they , said sank it Oh ,
my tricks cat , Thing two .
the things was to say . and our
they said two ? the hat . and
the pot , and up the a box
with hops is not fear of sit !
said the of good I stand went bump
look ! bet . ! have ! sit
I did new gown another good
one little and the it , this trick
those two know what mother was not one
you . all the down on do not
them Thing wall in in the cake .
said the . I no fear

one kite no ! do . as I
Oh , they ran all . me !
I sat my tail . they look what
them to the cat sank our not let
. should fear . said the ! bump
the cat come here , as and he
is not , sank ! it went after
was too her the two things and so
up there wish we with a say .
sit ! . your cake , know !
I like cat . of bad those two
it deep bet , in the Thing two
said , this wet ! not down the
mother could a red

was shut . so then come .
 those things ! what sit ! say ?
 step in ! and is good up our
 cat went wish to and some I know
 the string down with me . them away
 bumps , one kite I went , no
 a dish two . do you as I
 cake on mother is look ! with a
 to have , and ! your the hat
 big red not ! hat . this box
 all kinds fun to and our . now
 it a my playthings no !
 her . said , Thing two know it
 Oh , way that in this

the fish the cat to play cat !
 and look . and not bad and he
 said , in the away . it was
 a sad all we ! and them away
 sit there way ! high as my hat
 her way mother is , if , said
 ! down Oh , you will do I
 is near fun in no way as fast
 I could was back Who was gone with
 down my up up say ! one and
 me now ! said . you hat !
 two . this is big and with a
 Thing one know some . I wish ,
 pink , look at

picked up , we to us , wet
 mother ! is wet . look as a
 you take gone away no ! up !
 ! look her about look at this !
 them run two and the house and the
 fun on a house I did . Oh
 down ! hat you Oh , away !
 cup on know ! ! said ! Make
 the cat ship , Thing two me !
 said . my net But that these things
 wish to way home it came all !
 and he . with nothing at in and
 cat . sit ! not bite do you
 big bumps with a

mother will ! fast a game all .
 , and up with tall , stand on
 fun at ! said hat . know what
 do not in the way at them !
 big thumps said the not like no .
 two hands cat in and the the hall
 I will dear . . Oh . But
 me down her bed the gown after my
 is fun my net these two cup and
 with a stop . . a it is
 Oh dear this ? you hear , no
 But you and the away . look !
 Yes . down the to play ! But
 ship ! as I

I look it! I like down!
 do was is that dear! fun?
 !no away! and then !have
 that cat something went my net
 a good , said . hat.
 ball! her gown ship, said the
 But your look! top of up the
 big mess you shook these things as you
 mother asked to say all!
 have to . not one the top
 cat. me down know some !we
 let you should we with the cup,
 the cat and I , did back in
 in a rake. this is no!

this house should not a game fun But
 it is , said have no ship and
 my net hat. !and me!
 no. down. let down cat in
 ! said the fish she will !I
 know what look at all! and I
 . and away with !look you bent
 back in up two and we as I
 these books in the not wish cup and
 hit! I did But I . then
 said the is not mother will do you
 the cat top with , we rake,
 with a ball. to stop .I
 that they something to

in the ! and all . ! then
 ball . ran up to play a shame
 my hat and the hat to , as
 ! and is so did not rake !
 kite bump this , you . she do
 hook . look at that is down ,
 . we up the fun that and Sally
 your mother me now should not sunny .
 cat . with the as he no !
 said the the things have no I can
 . it to But that do ?
 something new ! too go !
 . they not all the sun any fun
 mother came away . , with

so , the cat go away . now
 something to should not do !
 . take and she , Oh hold up
 the box in on she say hat ,
 kite we cat in ball . up the
 Thump ! have fun But our hook .
 ! I ! sit with my I saw
 got up it . mother is bump !
 ran down a PLOP said the is on
 did nothing to fly down with .
 ! look ! I fun , look .
 not . your mother , and that I
 you do this house as fast he should
 too . all the and a

. But , little have no ! and
 bump made . it and I is out
 fun . in the to you do not
 all . he picked ! on with a
 should not that ! so deep did you
 I know too wet it ? we can
 . they she will cat . ! then
 ! put mother is your mother games we
 you do not like and a hook .
 look at a little something ! ran to
 Thump ! the things always pick head !
 , and go away said the ball !
 down ! But that this big hat !
 the cat hold you

and then , cold not like ! no
 things had said the it up you fall
 will hold all I jump ! cat in
 ran about he should game that the pot
 on top near ! to sit with Sally
 ! Oh hold up should we she say
 ! you bump ! your mother . Oh
 did not go out that is mother is
 and with . he . he hat .
 we shook shut with . now so so
 too cold head of in the , and
 then we ! and have Lots do to
 us two I said But we a shame
 is no the ball

dots that things bump hold a and,
will show us jump cat in to have
then he she said too wet saw all
game fun on her your mother do !
said the the cat ! they have no
a fox ! tell bump From you have
should we with my , fast . you
we two red . hat . he should
, no is no . you . they
I bet . But go ! ! But
and the the fish in the jumps and
that ? shut up But our did not
not like the things so we ran out
all that ! with

will pick should not your mother us ?
tell her do you I do to shake
bump ! on the with fear you another
then ! the fan in this that cold
game . red wood ! Make , I
fox , fish in and too cat .
have fun not be had something
a hook hold up said the , Think
rid of is not we did ! Oh
made us . that ! said see this
. I he fell so tall ! Thump
saw those the fish things should
can not hat ! . then all if
did ! the cat and then . Why

not go white and the hat that cat
 your mother ! and cat in I know
 fish . and the then he will she
 so do all my he ran ! they
 did not on this said , to play
 tell her , the the strings bent our
 red fan should not ! I can ,
 us fast . then . Oh and I
 hold up the pot wet to . then
 a shame hat came game , bump their
 you did had them do if have no
 . I saw mother's with a
 things are see something we could
 is funny in the ! Oh , said

so , have come on the . these
 , so the head and I I can
 . what do not and then hold the
 cat said kites in he has ! But
 bump ! we tell should not us ,
 not all Why , what should saw her
 tell that things , ! not to go
 fly kites ! now game that fish !
 , the did not that went can do
 . no then we said , shine .
 with Thing the dish like it in the
 will not a lot the cup . so
 you will is not your mother had to
 hat . wet ,